Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Based Clinical Trial Protocol Design System for Medical Devices Aligned with FDA Regulatory Requirements			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (v), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (v), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio・Medical		Knowledge Service	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	Medical Information Standardization	R&D / Engineering Services	Other R&D / Engineering-Related Technologies
Concept of Proposed Technology	The proposed technology is an **AI-based clinical trial design system for medical devices**, built on a **modular Retrieval-Augmented Generation (RAG) architecture**. The system is trained on **global regulatory guidelines** issued by the U.S. Food and Drug Administration (FDA), the European Medicines Agency (EMA), and other regulatory authorities, as well as **more than 700,000 historical clinical trial datasets**. When users input the **medical device classification and intended clinical indication**, the system automatically generates a **regulatory-aligned Clinical Trial Protocol (CTP) draft**. This process is designed to be completed **within three weeks**, representing an estimated **80% reduction in preparation time** compared to conventional manual protocol development workflows. The system is intended to support early-stage clinical planning by improving protocol structure, regulatory consistency, and operational clarity for medical device trials.			

Rationale for Joint R&D Project	This joint R&D project is structured as a **non-commercial, strategic research collaboration** that combines complementary strengths in **AI-driven protocol automation** and **U.S.-based regulatory and clinical trial expertise**. The collaboration is designed to build a **globally scalable, FDA-aligned AI validation framework** for medical device clinical trial design while preserving the independence and intellectual property of all participating parties. **(1) Translation of U.S. Regulatory Expertise into AI Validation** Medical device clinical trials in the United States operate within a highly experience-driven regulatory environment. Regulatory expectations are shaped not only by formal guidance documents but also by accumulated review practices and precedents. Through this joint research initiative, U.S.-based regulatory experience—derived from interactions such as IDE submissions, pre-submission consultations, and regulatory feedback—is applied to the **validation and refinement of AI-generated clinical trial protocols**. This enables systematic testing and improvement of the AI system without transferring proprietary methodologies or commercial rights. **(2) Institutional Experience for Research Validation** The overseas research partner contributes **de-identified historical clinical trial insights** exclusively for research validation purposes. These contributions support: • Comparative evaluation of AI-generated protocols against FDA-reviewed precedents • Identification of regulatory risk patterns and protocol design gaps • Enhancement of institutional understanding of AI-assisted trial planning Participation is limited to a **research collaborator role**, without licensing, commercialization, or ownership transfer. **(3) Advancement of Sponsor-Relevant Research Outcomes** Although the collaboration is non-commercial, the research outputs are aligned with sponsor needs by: • Reducing conceptual gaps between AI-generated protocols and regulatory expectations • Improving clarity and consistency during early-stage trial planning • Establishing evidence-based validation of AI-assisted clinical trial design methodologies The project establishes a foundation for future dialogue without presuming commercial, licensing, or revenue-sharing arrangements.
Key Words	1. Medical Device Clinical Trial 2. Generative AI (RAG) 3. Protocol Design Automation 4. FDA Regulatory Compliance 5. Digital Health

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (6)	End of the Project	Level (8)
R&D Objectives	• Validate the **accuracy and regulatory alignment** of AI-generated clinical trial protocols • Evaluate **efficiency gains** relative to conventional protocol design processes • Establish a research-validated AI framework for medical device clinical trial planning Design accuracy is validated through **comparative analysis** between AI-generated protocols and historical clinical trial data derived from U.S. research partner institutions.			
R&D Target Specifications	• **Clinical Trial Design Timeline:** Reduction from approximately 3 months to **within 3 weeks** • **Protocol Regulatory Compliance: ≥ 90% alignment** with FDA regulatory guidelines • **Cost Efficiency:** Approximately **50% reduction** in costs during the clinical preparation phase (preclinical to Phase I)			

R&D Plan	**Year 1: U.S. Localization and Data Integration**
	
	• **Korean SME Partner:**
	  ○ Construction of an FDA regulatory database
	  ○ Enhancement of the RAG-based AI architecture
	• **Overseas Research Partner:**
	  ○ Provision of U.S. clinical trial datasets
	  ○ Definition of regulatory and operational validation requirements
	
	Year 2: Joint Validation and Pilot Testing
	
	• **Joint Activities:**
	  ○ Evaluation of AI-generated protocol quality through investigator-initiated trials (IITs) or virtual clinical simulations
	• **Korean SME Partner:**
	  ○ UI/UX optimization based on researcher feedback
	  ○ Launch of a beta-version SaaS service for research validation			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Provide interpretation and advisory support on FDA regulatory expectations • Review medical and scientific validity of AI-generated clinical trial protocols • Execute usability testing with active researchers • Provide structured feedback for system refinement

Expected R&R of Korean SME Partner	**AI Platform Development and Operations** • Lead development of the AI-based clinical trial protocol design platform • Integrate global clinical trial datasets (700,000+) **Service Advancement and Scale-Up** • Advance platform based on joint research outcomes

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Outcomes • Establishment of a **first-in-class AI-based clinical trial design system** aligned with FDA and global regulatory standards • Creation of a core infrastructure technology supporting digital transformation of clinical trial planning Economic and Industrial Impact • Improved R&D productivity through reduced clinical preparation timelines • Reduction of failure-related costs in early-stage clinical development • Lower regulatory and partnership barriers for Korean SMEs entering U.S. medical device markets Commercialization Potential • Subscription-based SaaS model targeting academic medical centers, medical device companies, and CROs • Strengthened global credibility through joint Korea–U.S. research validation and publications

6. References

(1) Achieving Best-in-Class Clinical Trial Delivery: The Road to 2035 (McKinsey, 2025)

(2) J Am Med Inform Assoc . 2025 Mar 1;32(3):447-458.